Physiology of the Nervous system

Lecture (3)

Receptors. Sensory system (Part 1)

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Illustrations from (Lippincott illustrated medical physiology)

Introduction…

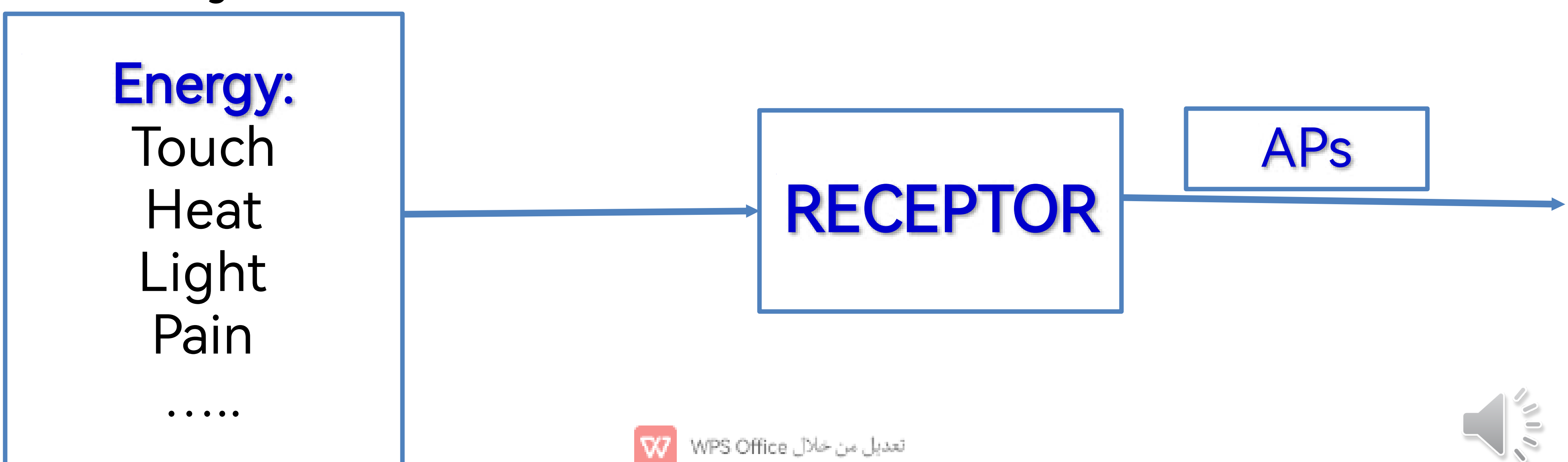
The function of the nervous system is to receive information about the internal and external environment through the sensory part, analyse the information and form an adequate response via the motor part

The receiving is called **reception**, and it is achieved by the receptors

The final translation of the sensory information is called **perception** by the cortex

Receptors as transducers…

Any device that is capable of converting one form of energy into a different (second) form of energy is called a receptor. In biology the (second) form of energy is always an **ACTION POTENTIAL**



The sensory system…

The sensory system is comprised of **4** components:

❖ Receptor (R)

❖ Afferent pathway **<u>(outside the CNS) (A)</u>**

❖ Ascending tract **<u>(within the CNS) (AT)</u>**

❖ Sensory cortex (SC)

R → **A** → **AT** → **SC**

Classification of receptors…

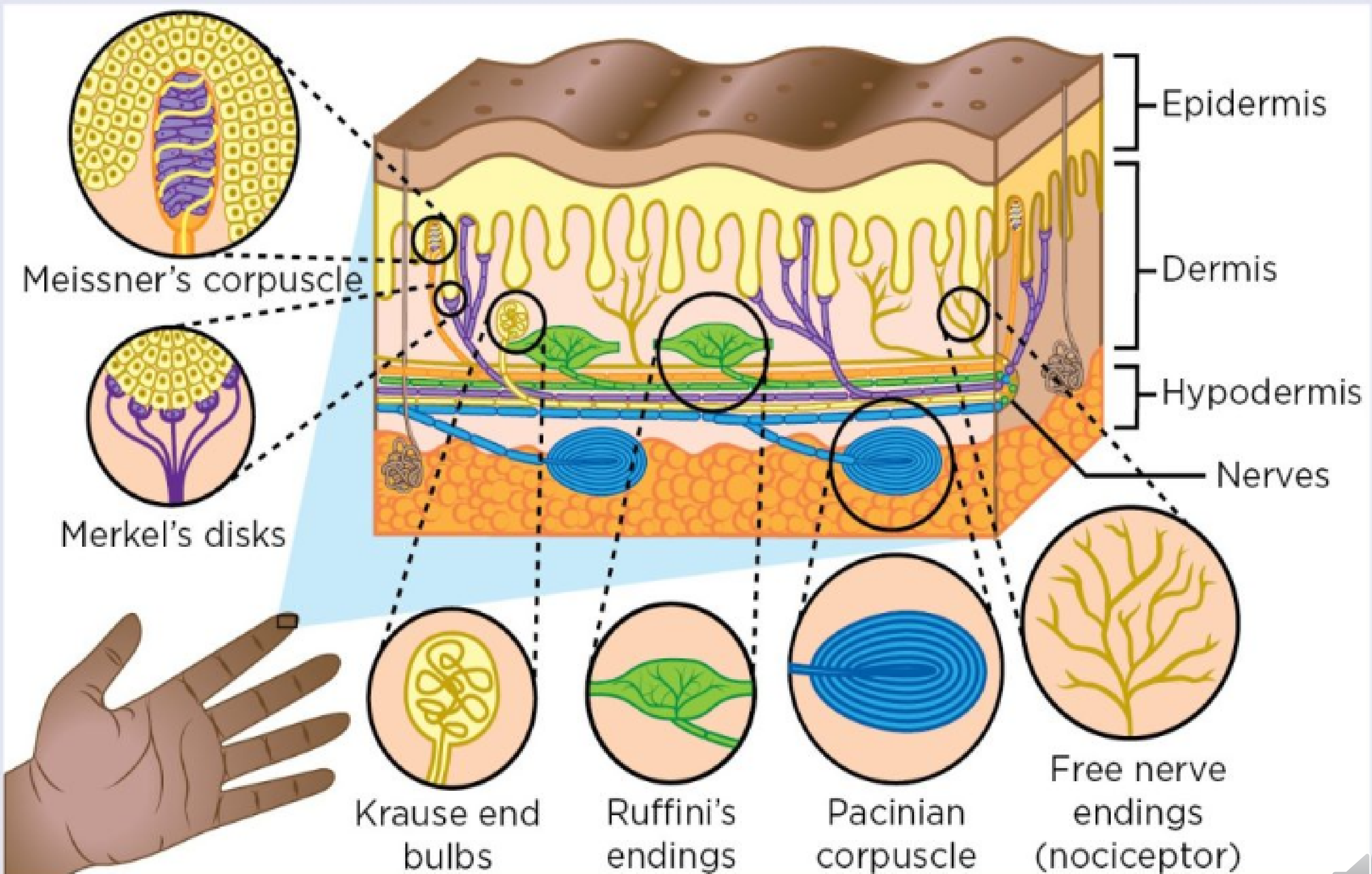
- According to location: (Extroceptors, interceptors)
- Teleceptors: from a distance (eye, ear)
- Nociceptors: respond to painful stimulus
- Mechnoceptors
- Chemoceptors
- Proprioceptors: in locomotor system
- Somatic vs visceral sensation

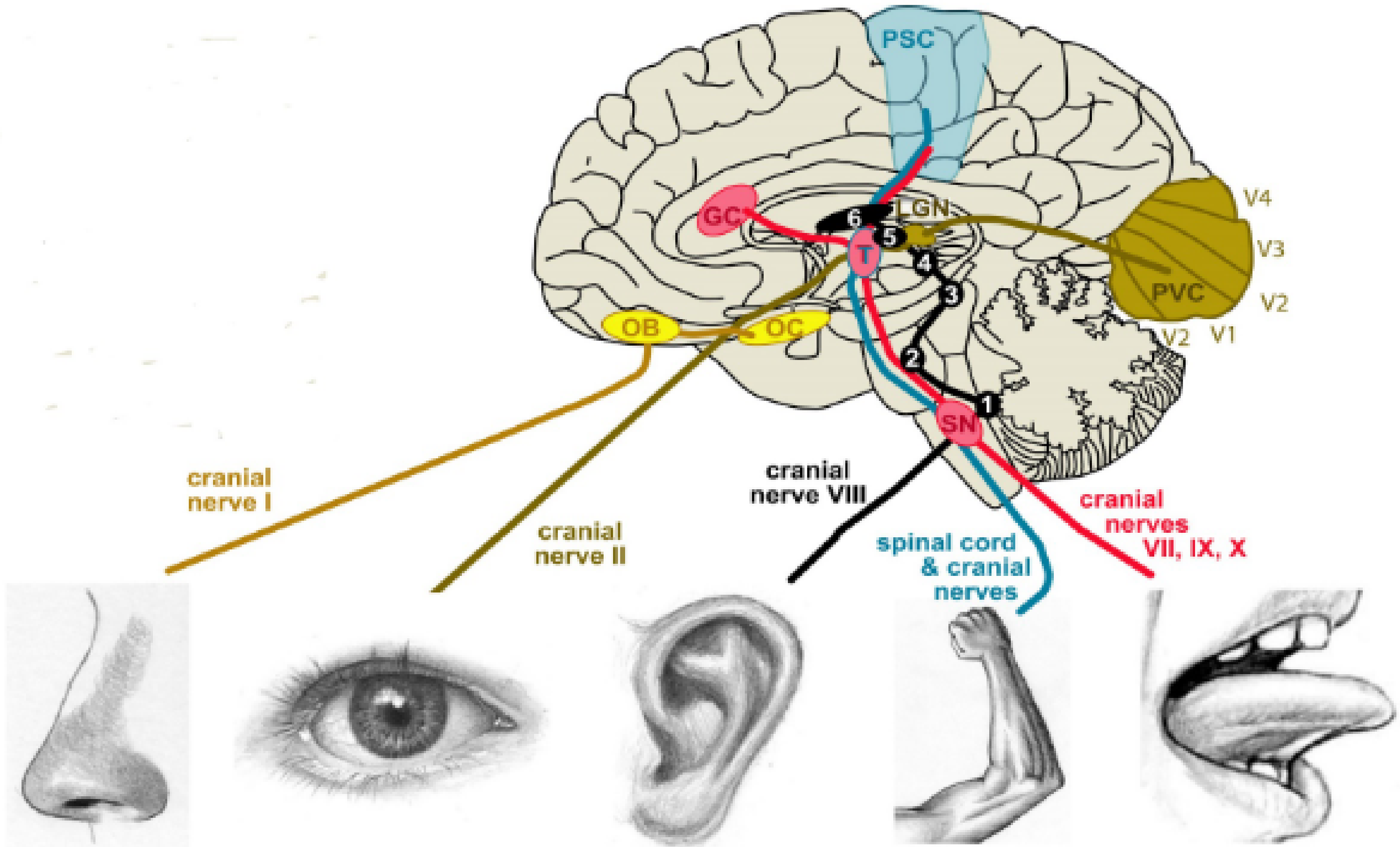
Receptors and their stimuli…

- Meissner’s corpuscle
- Markel's disk
- Krause blub
- Ruffini’s endings
- Pacinian corpuscle
- Free nerve ending

- Low frequency vibration
- Gentle touch
- Cold
- Stretch deformation
- High frequ. Vibration
- Tissue damage (pain)

Fig 1. **Mechanoreceptors in the somatic sensory system**





sense	Smell	Vision	Hearing	Touch	Taste
organ	**Nose** (nasal epithelium)	**Eye** (retina)	**Ear** (cochlea, inner ear)	**skin tongue**	**tongue, mouth wall, esophagus** (taste buds)
receptor cells	olfacotory receptor neurons	photoreceptor cells	mechano-receptive hair cells	mechano-receptive neurons	taste receptor cells
stimulus	chemicals	visible light	sound waves	mechanical forces	chemicals

Receptor properties…

Specificity (they respond to specific appropriate stimulus)

Excitability (should respond to a stimulus by generation potentials)

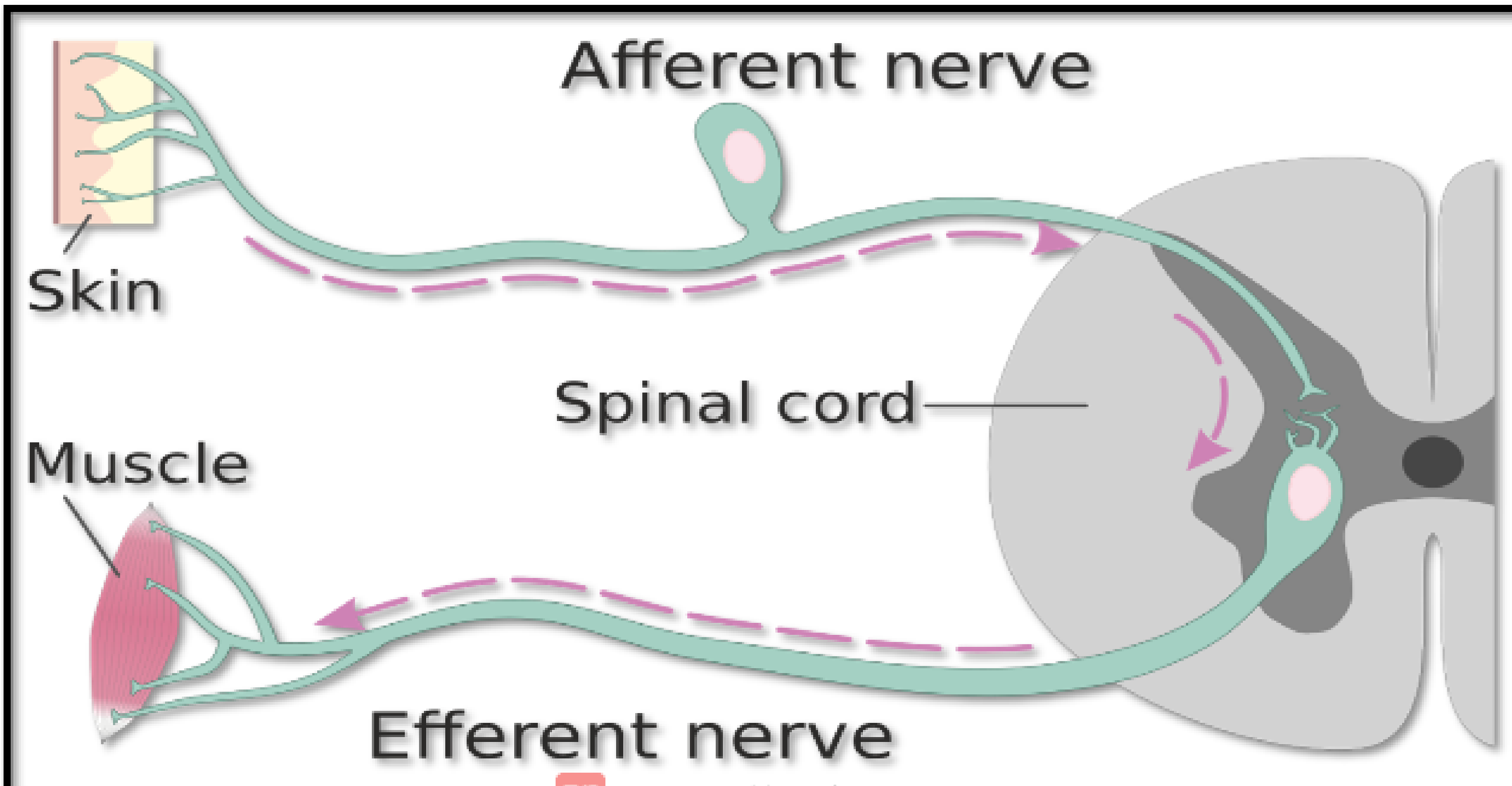
Adaptation they cease generating action potential after the stimulation

(rapidly adapting, slowly adapting, non-adapting)

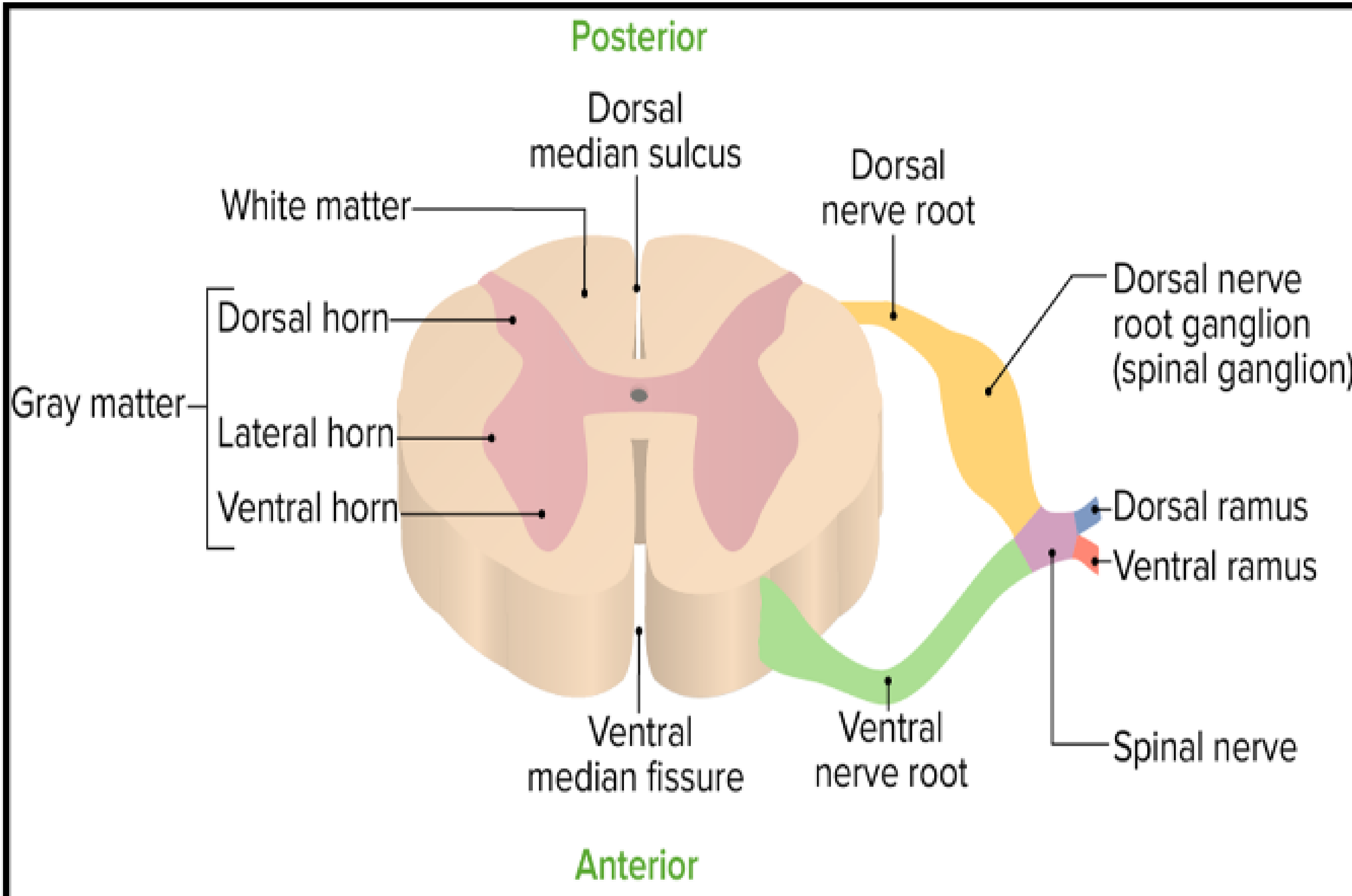
The generated potential at the level of the receptor is called receptor potential and it does not follow the all or none rule, has no propagation, however it can be summated to make an action potential

The afferent neuron…

- This is a sensory neuron, the axon of which is directing APs towards the CNS.



Spinal cord segment…



Neural connections within the CNS

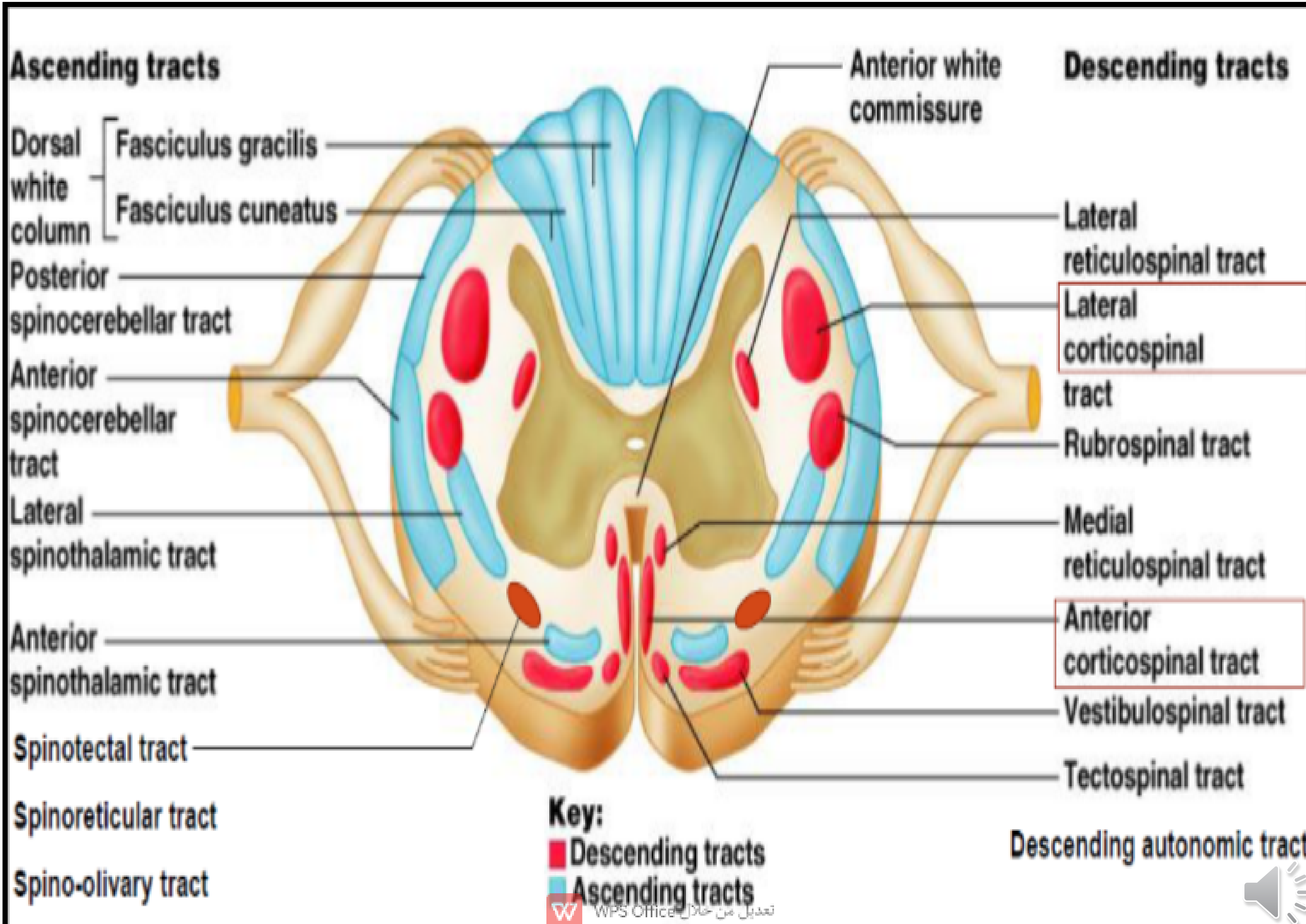
- ❖ Association fibers: bundles that connect regions within the same hemisphere
- ❖ Commissural fibers: bundles that connect the two hemispheres
- ❖ Projection fibers: bundles that connect each region to other part or to the spinal cord
- ❖ Tracts: A tract is a bundle of fibers that connects nuclei with the boundaries of the central nervous system, specifically between spinal cord and brain. They are:

Ascending: Sensory

Descending: Motor

Sensory pathways and tracts…

- Their function is to carry information from the receptors **(reception)** to the brain **(perception)** through a nerve in the periphery and a tract in the CNS
- They have 3 neurons:
- **1st** order neuron: from the receptor to the spinal segment
- **Second** order neuron: from the spinal segment to the thalamus
- **Third** order neuron: from the thalamus to the perception are in the cortex

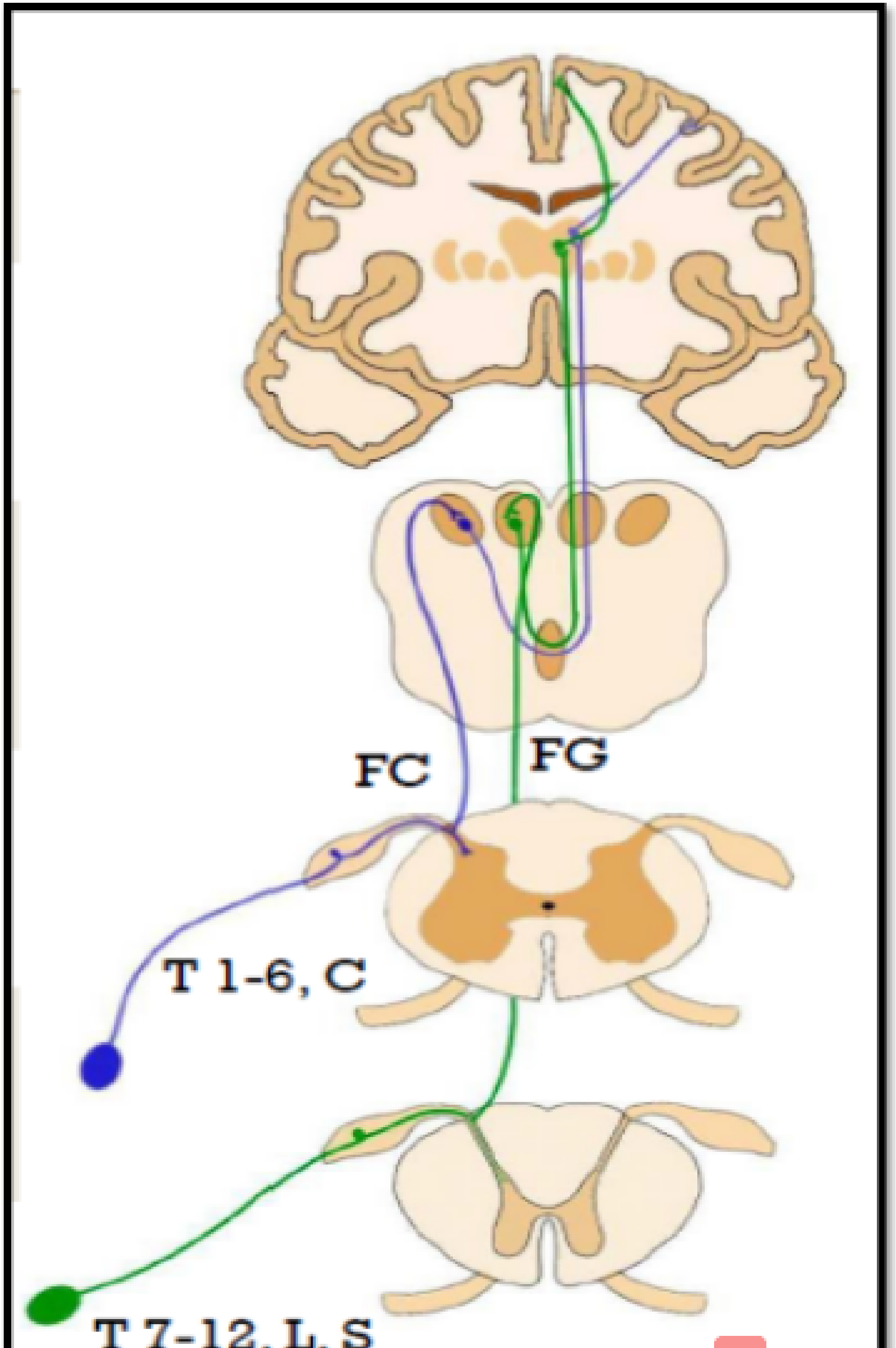

Ascending tracts
Dorsal white column
Fasciculus gracilis
Fasciculus cuneatus
Posterior spinocerebellar tract
Anterior spinocerebellar tract
Lateral spinothalamic tract
Anterior spinothalamic tract
Spinotectal tract
Spinoreticular tract
Spino-olivary tract
Anterior white commissure
Descending tracts
Lateral reticulospinal tract
Lateral corticospinal tract
Rubrospinal tract
Medial reticulospinal tract
Anterior corticospinal tract
Vestibulospinal tract
Tectospinal tract
Descending autonomic tract
Key:
Descending tracts
Ascending tracts

Sensation	Tract
Pain. Temperature	Lat. Spinothalamic
Crude touch. Pressure. Tickle and itch	Ant. Spinothalamic
Fine touch. Two point discrimination. Vibration Conscious proprioception	Dorsal column
Unconscious proprioception. Gross movement	Ant. Spinocerebellar
Unconscious proprioception. Fine movement	Pos. Spinocerebellar

Sensation	Tract
Proprioceptive information from Golgi organs, muscle spindles and joints	Spinocerebellar tracts (ant. Et Pos.)
Spinovisual reflexes, movement of the eye	Spinotectal
Pain perception	Spinoreticular
Cutaneous and proprioceptive information to the cerebellum	Spino-olivary
Hair movement, noxious	Spoinocervicothealmic

Dorsal column medial lemniscus tract...

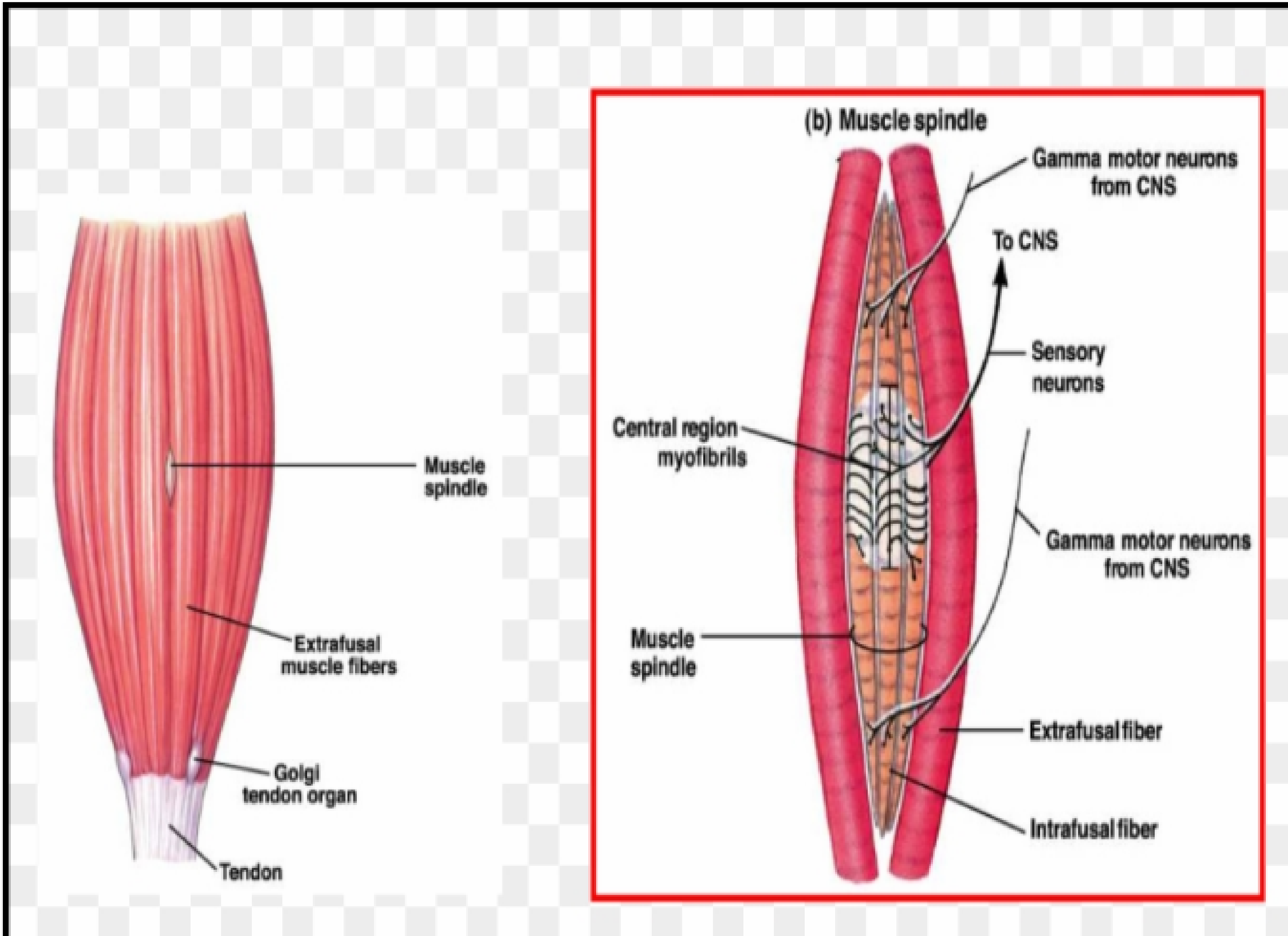
- (Fasciculus gracilis and cuneatus)
- Begins (receptors): Meissner's corpuscle, Pacinian corpuscle, muscle spindles and tendon organs
- Destination: Posterior central gyrus
- Carries fine and discriminative touch, pressure, stretch and vibration
- Proprioception **(knowing position in space)**



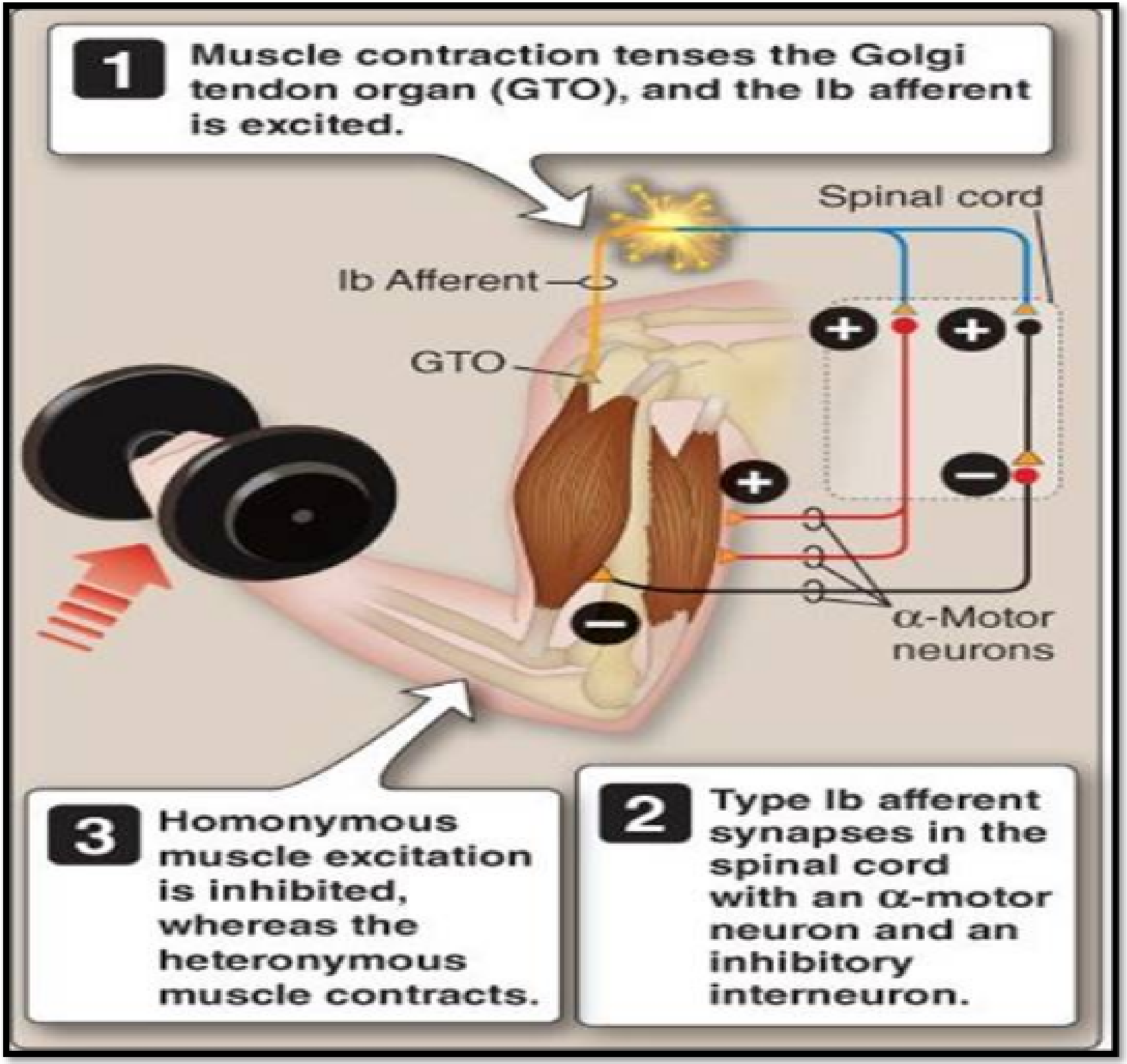
First order neuron:
Posterior root ganglion

Second order neuron:
Nuclei gracilis and cuneatus in the medulla oblongata

Third order neuron:
ventral posterior lateral nucleus of the thalamus



Muscle spindle
Extrafusal muscle fibers
Golgi tendon organ
Tendon
(b) Muscle spindle
Gamma motor neurons from CNS
To CNS
Sensory neurons
Central region myofibrils
Gamma motor neurons from CNS
Muscle spindle
Extrafusal fiber
Intrafusal fiber


1
Muscle contraction tenses the Golgi tendon organ (GTO), and the Ib afferent is excited.
Spinal cord
Ib Afferent
GTO
+
+
+
−
−
α-Motor neurons
3
Homonymous muscle excitation is inhibited, whereas the heteronymous muscle contracts.
2
Type Ib afferent synapses in the spinal cord with an α-motor neuron and an inhibitory interneuron.

Physiology lit. (recommended)…

- Guyton and Hall textbook of physiology
- Lippincott illustrated physiology
- Ganong review of medical physiology
- Linda Costanzo textbook of medical physiology
- BRS, Linda Costanzo
- Essential to physiology